

Literature Benchmark Brief: Ullah_2022_TII

Title: Enabling massive machine-type communications in offshore wind farms: A wireless connectivity perspective

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Cluster: Cluster B: SCADA Degradation & Reliability

Core Theoretical Mechanism & Model Formulation

LoRaWAN and LEO satellite uplink queuing latency and packet drop modeling in offshore wind

Empirical Scale & Experimental Dataset

Offshore wind farm communication network topologies

Relevance to IEEE TSTE & Sustainable Energy Systems

Establishes industrial inevitability of 10-60 min latency buffers and packet dropouts

Identified Gaps & Operational Vulnerabilities

Pure telecommunication focus without wind power dispatch consequence analysis

Role in Manuscript Defense & Reviewer #2 Proof

Defends engineering realism of Delay-1 to Delay-6 and Markov burst packet loss settings